



Fig. 2: Block diagram of a simple LCD controller. The blocks with white outlines represent some possible functions that can be implemented in FPGA devices. These functions can be either individually addressed as needed by smaller low-power reprogrammable FPGA devices or integrated into larger FPGA devices. These ultra-low-power FPGA families offer gate capacity from 10,000 gates up to 3 million gates.

where increasing emphasis is being placed on new and advanced technologies for prevention, early diagnosis and treatment.

Pressures to reduce healthcare costs are butting into the demand for more portable devices, which, when deployed in the homes, doctors' offices and elsewhere, can save the money spent on the healthcare system. Portable medical equipment made big news at the 2008 *Beijing Olympics* when portable MRIs and other imaging and diagnostic tools from companies like GE were used. These devices need to be low-power, flexible and cost-effective for wide deployment.

FPGAs—with core voltages down to 1.2V—play a big part in this development, allowing the programming of various features and design for differing geographical standards, while at the same time keeping power consumption low and design costs down.

Today, medical devices—for example, digital blood pressure meters, blood gas meters and blood glucose meters—must not only be ubiquitous but robust also. These are expected to do much more than just testing and monitoring. Some are now capable of logging and analysing data as well

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as communicating accurate results to the health provider. Blood pressure meters are now benefiting from a more extensive data logging feature as well as communication ports for real-time sharing of information with the health provider. Insulin meters are now equipped with communication ports (IR/wireless) to transfer real-time measurement to the PC or to the insulin pump to treat the disease. In their most basic form, these portable medical devices are all battery-operated, microcontrolled handheld devices with power constraints.

The insulin pump is typically controlled by a microprocessor. These microprocessors perform various functions, such as processing data from bio-sensors, storing measurements and analysing results. Gone are the days when the design tradeoff necessitated use of a slow, mokey mi-

croprocessor unit (MPU). Today, low-power designs can take advantage of full-featured processors, such as an FPGA-optimised 32-bit ARM Cortex-M1 processor.

But implementing MPU or controller features in a low-power FPGA is just one aspect of the evolving portable medical device design. One of the most important features—and potentially power-hungry—is the display.

Lower costs and ease of mass manufacturing have increased the demand for LCD panels in various home-based and commercial medical markets. When creating these devices to meet consumer demands, designers select LCD panels based on critical factors such as size, resolution, reliability, power consumption and product lifecycle.

As newer displays with enhanced capabilities and features are continuously being launched, designers face challenges in keeping up with technology by redesigning the display controller. Redesigning is expensive and can significantly increase time-to-market. Therefore designers need solutions that will enable them to incorporate the latest technology with minimal cost and effort. Ultra-low-power and reprogrammable FPGAs provide solutions to address these challenges.

In a nutshell

For three decades, FPGAs have carved out a profitable niche as customisable devices that give designers a choice over ASICs, especially when they want to get to market quickly in low volumes. But engineers who wanted to choose FPGAs for power-sensitive designs found themselves in a bind—until now. In an era when more devices are mobile and require long battery life, or when energy efficiency needs to be designed into new systems, flash-based FPGAs are emerging as a fast, flexible and affordable low-power alternative to SRAM-based FPGAs, ASICs and CPLDs. ●

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